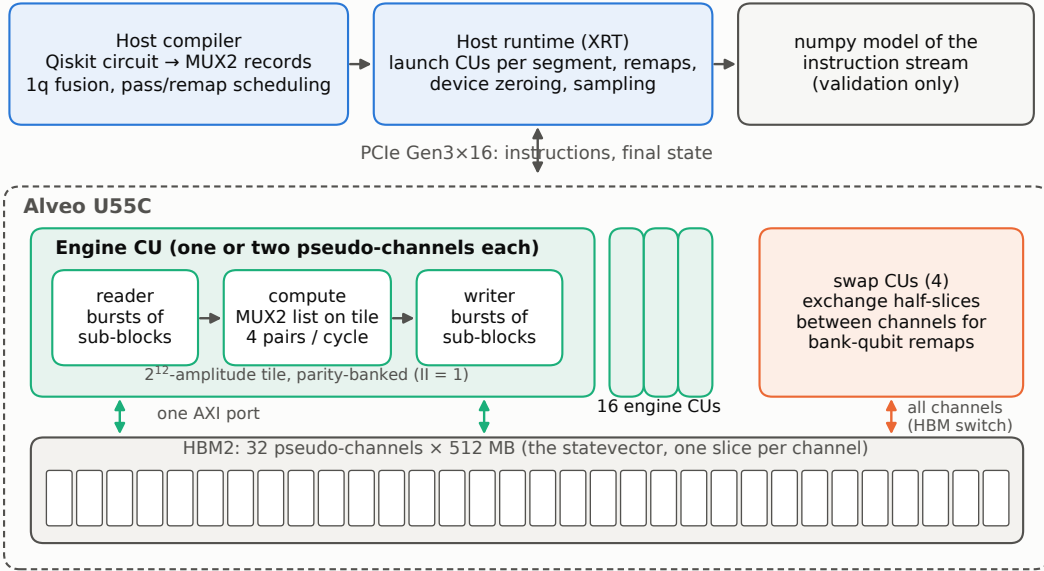
Host compiler
Qiskit circuit → MUX2 records
1q fusion, pass/remap scheduling
Host runtime (XRT)
launch CUs per segment, remaps,
device zeroing, sampling
numpy model of the
instruction stream
(validation only)
PCIe Gen3×16: instructions, final state
Alveo U55C
Engine CU (one or two pseudo-channels each)
reader
bursts of
sub-blocks
compute
MUX2 list on tile
4 pairs / cycle
writer
bursts of
sub-blocks
2^12-amplitude tile, parity-banked (II = 1)
16 engine CUs
swap CUs (4)
exchange half-slices
between channels for
bank-qubit remaps
one AXI port
all channels
(HBM switch)
HBM2: 32 pseudo-channels × 512 MB (the statevector, one slice per channel)